INVERTIS UNIVERSITY, BAREILLY

FACULTY OF ENGINEERING & TECHNOLOGY

OFFICIAL ACADEMIC LEARNING RESOURCE â¢ DEPARTMENT OF CSE

Course: [MCS-101] Advanced Computer Architecture

Programme: Master of Technology in Computer Science & Engineering **Resource Type: PPT SLIDES**

Module / Unit: Unit 1 **Faculty Author: Dr. A. K. Ganguly (Head of De**

Lecture Slides: Instruction-Level Parallelism, Speculation & Cache Coherence

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Slide 1: Pipeline Hazards & Dynamic Scheduling

- Hazards: Structural, Data (RAW, WAR, WAW), and Control branching hazards.
- Scoreboarding vs Tomasulo’s Algorithm: Register renaming and reservation stations.
- Reorder Buffer (ROB): In-order commit with out-of-order execution pipeline.

Slide 2: Multiprocessor Memory Architectures & Coherence

- Memory Consistency Models: Sequential consistency, relaxed consistency, release consistency.
- Cache Coherence Protocols: Snooping (MSI, MESI, MOESI) vs Directory-based protocols.
- Interconnection Networks: Crossbar switches, 2D mesh, torus, and hypercube topologies.